End-to-End Synthetic Song Detection with Transformer-Based Models

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Abstract

Recent text-to-song systems can generate full tracks with synthetic vocals, music, style, and lyrics, making counterfeit-song detection substantially harder than prior singing-voice deepfake settings. This paper presents a concise study of end-to-end synthetic song detection using the same dataset protocol and task setup as the primary benchmark paper in our references. We focus on a direct comparison of three transformer-based model families: ViT, EfficientViT, and Spectro-Temporal Tokens Transformer (SpecTTTra). Our study emphasizes long-context evaluation and reports model behavior in terms of detection performance and computational efficiency.

Keywords: AI-generated music detection, counterfeit song detection, transformer models, ViT, EfficientViT, SpecTTTra

1 Introduction

Modern music generation systems can synthesize complete songs end-to-end. This shifts the detection task from voice-only spoof detection to full-song authenticity verification. In this work, we study transformer-based detectors for this setting and compare their practical strengths and weaknesses.

Songs contain long-range structure (repeated sections, phrase-level evolution, rhythm, and timbral continuity), so long-context modeling is central to robust detection. We therefore evaluate models that can process spectro-temporal information at broader time scales.

2 Task Definition

We address binary *Synthetic Song Detection* (SSD): classify an input song as human-composed (real) or AI-generated (fake). Evaluation includes:

- seen vs. unseen generation algorithms,
- short vs. long audio clips,
- fake-intensity subsets (e.g., half fake, mostly fake, and full fake partitions).

3 Dataset

We use the same dataset and split protocol as the main benchmark reference listed in Section *References*. The dataset contains:

- 97,164 total songs,
- 49,074 synthetic songs,
- 48,090 real songs,
- 4,751 total hours,
- average song length of about 176 seconds.

This scale and duration support long-context analysis for song-level authenticity detection.

The dataset also includes structured metadata attributes in two CSV files:

- **real_songs.csv**: `id` (unique file ID), `filename` (file name), `title` (song title), `artist` (artist name), `year` (release year), `lyrics` (song lyrics), `lyrics_features` (LLM-extracted lyric features), `duration` (total seconds), `youtube_id` (source YouTube ID for real song), `label` (always real), `artist_overlap` (whether train/test share artist identity), `target` (0 for real), `skip_time` (instrumental-only time before vocals), `no_vocal` (True/False vocal presence), and `split` (train/valid/test).
- **fake_songs.csv**: `id` (unique file ID), `filename` (file name), `title` (song title), `duration` (total seconds), `algorithm` (generation algorithm), `style` (style characteristics), `source` (generated by Suno or Udio), `lyrics_features` (LLM-extracted lyric features), `topic` (song theme), `genre` (music genre), `mood` (song mood), `label` (full fake/half fake/mostly fake), `target` (1 for fake), and `split` (train/valid/test).

These attributes enable controlled analysis by generator type, style, lyrical content, vocal presence, and fake-intensity level.

[0]Code & Data available at: https://github.com/yuanmingwang/ai-music-detector-transformer

4 Models

We implement and compare three transformer-based model families:

- **ViT:** a vanilla vision-transformer style backbone adapted to spectrogram inputs.
- **EfficientViT:** a lightweight transformer design for improved speed/memory efficiency.
- **SpecTTTra:** a spectro-temporal token transformer designed for efficient long-context modeling.

For completeness, the core Transformer computations used in these architectures are:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{1}$$

$$\mathrm{MHA}(X) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h) W^O, \tag{2}$$

$$\mathrm{head}_i = \mathrm{Attention}(XW_i^Q, XW_i^K, XW_i^V), \tag{3}$$

$$\mathrm{FFN}(x) = W_2\, \sigma(W_1 x + b_1) + b_2. \tag{4}$$

One encoder layer is then written as:

$$Z' = \mathrm{LayerNorm}(Z + \mathrm{Dropout}(\mathrm{MHA}(Z))), \tag{5}$$

$$Z^{\mathrm{out}} = \mathrm{LayerNorm}(Z' + \mathrm{Dropout}(\mathrm{FFN}(Z'))). \tag{6}$$

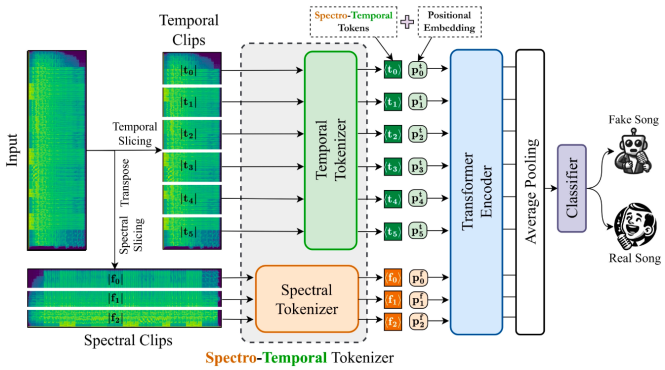

Figure 1: Proposed Spectro-Temporal Tokens Transformer (SpecTTTra) model.

All models output a song-level fake probability and are evaluated under the same data protocol for fair comparison.

5 Experimental Protocol

To ensure comparability, we use a shared training/evaluation setup across all models:

- identical train/validation/test partitions,
- consistent real/fake label definitions,
- F1 score as the primary metric,
- sensitivity and specificity as supporting metrics,
- runtime and memory usage for efficiency analysis.

6 Results Summary

6.1 Status Report — Suchang

Objective 1: Spectral-Artifact Detection Module

6.1.1 Overview

Work on the Spectral-Artifact Detection Module is progressing steadily. Core components of the pipeline—including dataset preparation, spectral feature extraction, and initial model prototyping—have been partially implemented and validated on a small-scale dataset. While full-scale training is pending due to ongoing dataset acquisition, early results indicate that the proposed approach is viable.

6.1.2 Progress Summary

Dataset Assembly

- Successfully downloaded initial portions of M6, FakeMusicCaps, and FMA.
- Due to the large size of SONICS and M6, implemented a partial-download strategy to enable early experimentation.
- Drafted a pipeline for collecting Suno/Udio-generated samples, currently pending confirmation of usage permissions.
- **Status:** Partially completed.
- **Next step:** Complete full dataset acquisition and unify formatting across datasets.

Literature Review

- Reviewed Fourier-domain artifact detection (Afchar et al.).
- Reviewed structural modeling via Segment Transformers (Kim & Go).
- Reviewed SSL-based representations (CLAP, wav2vec2, MERT).
- Reviewed explainability and multimodal detection methods.
- **Status:** On track.
- **Next step:** Integrate remaining papers as needed during model refinement.

Preprocessing & Feature Extraction

- Implemented multi-resolution STFT extraction (short, medium, long windows).
- Implemented basic beat tracking (Librosa).
- Implemented SSL embedding extraction (CLAP, wav2vec2).
- Implemented a prototype of Fourier-domain artifact fingerprints based on Afchar et al.

- All components were validated on a small dataset and produce expected outputs.
- **Status:** Core functionality implemented.
- **Next step:** Scale to full dataset and optimize computational efficiency.

Early Model Prototyping

- Trained a preliminary classifier using CLAP embeddings and Fourier-domain spectral fingerprints.
- Used a small curated dataset (AI-generated vs. human music).
- The model demonstrates clear separation between classes, suggesting spectral artifacts are informative features.
- Current limitations: dataset size and unevaluated robustness under pitch-shift, time-stretch, and compression.
- **Status:** Prototype completed.
- **Next step:** Retrain with full dataset and conduct robustness testing.

6.1.3 Alignment with Design Objectives

- PI1: Multi-resolution STFT — Completed.
- PI2: Fourier peak detection — Partially implemented.
- PI3: Artifact fingerprints (Afchar et al.) — Prototype implemented.
- PI4: Classifier with spectral + SSL features — Prototype completed.
- PI5: Robust, generator-agnostic detector — Not yet started.

6.1.4 Remaining Work Plan

Dataset Scaling

- Complete downloads of SONICS and M6.
- Finalize Suno/Udio dataset acquisition.
- Standardize preprocessing pipeline across datasets.

Model Improvement

- Train on full-scale dataset.
- Improve classifier architecture and feature fusion.
- Conduct robustness testing under pitch shifting, time stretching, and noise/codec compression.

Advanced Objective

- Develop a generator-agnostic artifact detector.
- Evaluate generalization on unseen generators.

6.1.5 Current Risks & Challenges

- Dataset size and availability are currently the main bottleneck.
- Usage permissions for some generated datasets (Suno/Udio) remain unresolved.
- Computational cost may increase significantly when scaling to full datasets.

6.1.6 Conclusion

The project is progressing according to plan. Core modules have been implemented and validated at a small scale, and initial results confirm the feasibility of spectral artifact-based detection. The next phase will focus on dataset scaling, robustness evaluation, and improving generalization performance.

6.2 Status Report — Yuanming Wang

6.2.1 Final Project Goals

The final goal of this project is to develop a reliable system for detecting AI-generated music by leveraging modern deep learning techniques, with a particular focus on transformer-based architectures.

Instead of implementing a complex multi-layer detection pipeline, the project adopts a focused and experimental approach by comparing multiple state-of-the-art models. Specifically, we aim to design, implement, and evaluate three different transformer-based models for binary classification between real and AI-generated music.

The project will proceed through the following stages:

Dataset Preparation

- Construct a labeled dataset consisting of real songs (collected from online sources such as YouTube) and AI-generated music samples.
- Ensure diversity in genre, quality, and duration.

Feature Extraction

- Convert raw audio into suitable representations such as Mel-spectrograms or learned embeddings.
- Standardize preprocessing for consistent model input.

Model Development

- Implement and compare three transformer-based architectures:
- A baseline transformer model.
- A hybrid CNN-transformer model.
- An advanced transformer-based audio classifier.

Training and Evaluation

- Train all models under consistent conditions.
- Evaluate performance using metrics such as accuracy, precision, recall, and F1-score.

Comparative Analysis

- Analyze differences between models.
- Identify which architecture is most effective for AI music detection.

The final outcome will be a comprehensive evaluation of transformer-based approaches for detecting AI-generated music, along with insights into their strengths and limitations.

6.2.2 Progress Toward Objectives

Dataset construction and curation (major progress).

- Built a labeled dataset containing real songs and AI-generated songs.
- Organized data into a structured storage format for downstream training.

Data acquisition pipeline (core functionality completed).

- Implemented an automated script-based pipeline to download real songs from metadata (e.g., YouTube IDs).
- Standardized folder-level organization for reproducible preprocessing.

Literature review and methodology refinement (strong progress).

- Reviewed recent work on AI-generated music detection and transformer-based modeling.
- Refined project scope from a broad multi-layer system to a focused three-model transformer comparison.

Modeling and feature pipeline (in progress).

- Model families and experimental protocol are defined.
- Feature extraction implementation and full training runs are the next milestones.

6.2.3 Performance Indicators Status

- **Dataset collection (real + AI songs):** Completed.
- **Automated data pipeline:** Completed.
- **Literature review and direction definition:** Completed.
- **Model architecture design:** In progress.
- **Feature extraction pipeline:** Not started.
- **Model training and evaluation framework:** Not started.

6.2.4 Yuanming Wang's Individual Contribution

- Designed and organized the dataset structure for real and AI-generated music.
- Conducted data collection for AI-generated fake songs.
- Implemented the data collection and ingestion workflow for real-song acquisition.
- Conducted literature review and proposed the three-transformer comparative direction.
- Helped transition the project from a generic multi-layer concept to a feasible model-driven experimental framework.

6.2.5 Challenges Encountered

- Data collection instability from online sources (e.g., download failures and source-side restrictions).
- Maintaining class balance and diversity across genre, quality, and clip duration.
- Scope control: narrowing an initially broad design into a tractable and measurable experimental plan.

7 Discussion

These results support using transformer-based long-context modeling for modern counterfeit-song detection. In particular, efficiency-aware architectures are important for real-world use cases such as moderation pipelines, forensic screening, and large-scale catalog analysis.

Future work should further improve robustness under unseen generators and manipulation attacks, and can combine audio models with lyrics, metadata, and production-level cues.

8 Conclusion

We presented a concise comparison of ViT, EfficientViT, and SpecTTTra for end-to-end synthetic song detection using a shared large-scale benchmark protocol. The study highlights that architecture choice matters not only for F1 performance, but also for practical efficiency on long audio.

Acknowledgments

This paper is derived from the benchmark setting and related prior work listed in the references.

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